BPG-11 · BEST PRACTICE GUIDES

Change orders and variations

A change to the work and a change to the baseline are two separate events

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PROJECT CONTROLS INSTITUTE GLOBAL, INC.

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Summary

A change to the work and a change to the baseline are two separate events with different authorities and different dates, and the gap between them is where control is lost. This guide covers who may instruct a change, why notice provisions decide entitlement before anyone argues about value, the pricing bases and what each does to productivity risk, how instructed-but-unpriced and disputed variations should sit in a forecast, why a variation dropped into the baseline at the wrong date corrupts the performance indices, and the log discipline that holds it together.

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In one paragraph. A change to the work and a change to the baseline are two separate events with different authorities and different dates, and the gap between them is where control is lost. This guide covers who may instruct a change, why notice provisions decide entitlement before anyone argues about value, the pricing bases and what each does to productivity risk, how instructed-but-unpriced and disputed variations should sit in a forecast, why a variation dropped into the baseline at the wrong date corrupts the performance indices, and the log discipline that holds it together.

Who this is for. Quantity surveyors, commercial managers, cost engineers and project controls managers on either side of a contract, and the project managers who sign the instructions.

This guide is not legal advice. Entitlement to additional payment or time, the effect of failing to give notice, and the valuation rules that apply all depend on the specific contract and on the law governing it. Nothing here describes the position under any particular jurisdiction or standard form. Take advice on your own contract.

1. Two changes, two clocks

When a client instructs additional work, two distinct things need to happen and they almost never happen together:

1. **The scope changes.** From the moment of instruction, the work required is different from the work priced. Resources are committed and cost begins to accrue.
2. **The baseline changes.** At some later point — after the value is agreed and a change order issued — the budget, the schedule and the measurement yardstick are updated to include the new work.

The interval between them is often months. Inside it, the project performs work its baseline does not recognise, incurs cost its budget does not contain, and reports indices that measure new work against an old plan. Nothing is wrong with the project; the reporting is describing a project that no longer exists.

Three cases are worth separating, because they are governed differently:

- **Scope change with a baseline change** — the ordinary, well-run case.
- **Scope change without (yet) a baseline change** — the instructed but unpriced variation. Legitimate, temporary, and dangerous if not disclosed.
- **Baseline change without a scope change** — re-phasing, transfers between control accounts, contingency drawdown. These are internal and must never be described as variations, because they carry no commercial entitlement.

The controls consequence: a change register that only records agreed change orders is recording the second event and missing the first. The register must start at the instruction, not at the agreement.

2. Instruction, authority and the notice trap

Authority. Contracts name who may instruct a change. Work performed on the say-so of someone without that authority is, commercially, work performed for free until proved otherwise. This is not a theoretical risk: it is the single most common source of unrecovered cost on a busy site, because the instruction usually arrives as a reasonable-sounding request from someone senior standing next to the work.

The practical control is unglamorous and effective: **no change is worked until it has an instruction reference in the log.** Where site conditions make that impossible, the person who accepted the direction records it the same day, in writing, to the named contract representative, describing what was directed and by whom.

Notice. Most contracts require the party seeking additional time or money to notify within a stated period of the event, or of becoming aware of it. What happens if that period is missed varies enormously: under some contracts and some governing laws the failure bars the entitlement outright; under others it affects only the evidence or the quantum; under others still it has little effect. Standard forms differ from each other, and the same form can operate differently under different governing law. **Never assume the position on your project — read the contract and take advice.**

What is universal is the control. Notice periods are short, they run from an event date rather than from the date somebody noticed, and they are missed for administrative reasons rather than commercial ones. A project therefore needs:

- **A notice diary** with one owner, listing every potential entitlement event, its event date, the applicable period and the due date.
- **A weekly review** of items approaching their due date, at a meeting where somebody can actually issue the notice.
- **A default to notify.** The cost of an unnecessary notice is a small amount of goodwill. The cost of a missed one can be the whole claim.

Change by conduct. Work can also be changed without an instruction — by a drawing revision, by a specification clarification that clarifies nothing, by acts that alter the basis on which the work was priced. Whether that carries entitlement depends entirely on the contract and the governing law; the controls discipline does not change. Record it as a potential change on the day it is noticed, price it, and give notice.

3. Pricing bases and who carries the productivity risk

Contracts set out how varied work is valued, and the hierarchy typically moves from "use the rates you already agreed" towards "price it afresh" as the work departs further from what was priced. Four bases cover most situations.

Contract or bill rates. Where the varied work is of similar character and executed under similar conditions to work already priced, the existing rates apply. Fast, uncontentious, and it leaves productivity risk where the original contract put it.

Pro-rata or "star" rates. Where the work is similar in kind but different in conditions — higher, deeper, smaller quantities, worse access — a rate is derived from the contract rates and adjusted. The adjustment is the argument; the derivation should be shown element by element.

A new rate built up from first principles. Where the work bears no useful relation to anything priced, the rate is constructed from materials, labour hours and rate, plant, on-costs and profit. Agreed before work starts, it becomes a lump sum and the contractor carries the productivity risk.

Daywork or cost reimbursement with defined additions. The work is recorded as it happens and reimbursed at daywork rates or at cost plus a stated percentage. This moves productivity risk to the paying party, which is why paying parties resist it and why daywork records must be signed every day by someone with authority. An unsigned sheet produced three months later is worth very little.

Two points that save money:

- **The basis is determined by the contract, not by preference.** Choosing daywork because it is easier to administer, or bill rates because they are favourable, is not valuation; it is negotiation dressed as valuation, and it collapses under scrutiny.
- **Time-related cost is a separate question from the direct works.** A variation that extends the duration of the works may attract prolongation cost, and that is a different assessment with different evidence. Pricing the direct works and assuming the time cost is "in the rates" is how contractors lose the second half of a variation. Where a delay is involved, see `BPG-12 — Claims and extension of time`.

4. Where variations sit in a forecast

At any data date a project's change population falls into three buckets, and each is treated differently.

Bucket 1 — instructed and agreed. The value is fixed and a change order has been issued. The budget is added to the baseline, time-phased to when the work is planned (see §5), and the work then earns value normally. Both cost and recovery are in the numbers.

Bucket 2 — instructed but not yet priced or agreed. The work is proceeding; the value is not settled. The cost belongs in the forecast at the best current estimate, because it is going to be incurred. The budget does **not** belong in the baseline, because there is no agreed value to add. The result is an account whose estimate at completion legitimately exceeds its budget at completion, and the report must say why — otherwise a reader sees an overrun where there is extra scope.

Bucket 3 — notified, claimed or disputed. Entitlement is asserted but not accepted. Cost still belongs in the forecast where it will be incurred. Recovery is the difficult half, and the discipline is asymmetric: **recognise the cost when it is probable; recognise the recovery only when it meets the applicable recognition threshold.** Revenue standards work on this principle — IFRS 15, for example, constrains variable consideration such as a claimed variation — but the exact threshold, and the treatment in management accounts as opposed to statutory ones, depends on the reporting framework in use. Take the accounting position from the finance function; take the control position from this rule: **never forecast cost net of an assumed recovery.**

Report the estimate at completion in two lines: the forecast for the baselined scope, and the forecast effect of unincorporated change. A single blended number hides which of the two is moving, and they move for entirely different reasons.

5. Time-phasing a variation into the baseline

When an agreed variation enters the baseline, its budget must be spread across the periods in which the varied work is planned to be performed. This sounds obvious and is routinely got wrong, because the simplest thing a cost system can do is add the budget at the date of the change order.

The effect is immediate and entirely artificial. Adding planned value in the current period for work not yet started leaves earned value unchanged and the denominator of the schedule performance index larger, so the index falls. Nothing about performance has changed; the project has simply been credited with a plan it was never given a chance to meet. §9 quantifies the effect: an index drop of 0.030 created by a filing decision.

The same distortion runs the other way at the end. A variation phased into months that have already passed inflates planned value retrospectively and creates schedule variance that no action can recover.

The rule: **the variation's budget is phased to the varied work's planned dates, and the schedule is updated in the same cycle.** If the varied work has not been programmed, the variation is not ready to be incorporated — and that, not the value, is usually the thing holding it up.

6. The change order log

One log, one owner, reviewed weekly. It is the project's commercial memory, and it is also — although nobody calls it this at the time — the first draft of any claim file that may be needed later. The columns that earn their place:

Column	Why it is there
Change identifier	The permanent citation key; never reused
Event date	When the change arose — the date notice periods run from
Notice date and reference	Evidence that notice was given, and when
Instruction reference and issuer	Establishes authority (§2)
Description and originating cause	Client change, design development, site condition, error
Bucket (agreed / instructed-unpriced / claimed)	Drives the forecast treatment (§4)
Pricing basis	Bill rates, star rate, quotation, daywork (§3)
Submitted value	What was asked for
Assessed or certified value	What was accepted
Time effect claimed and granted, in days	Kept separate from money, always
Affected control accounts and cost codes	So cost can be isolated as it is incurred
Status, next action, owner, next action date	Turns a register into a live control
Age in current status	The single most useful column in the log

Ageing is the control. Set a threshold — nothing sits in "submitted, awaiting assessment" for more than a stated number of weeks — and report the ageing profile monthly alongside the values. A log where the average age of unpriced variations is rising is a project whose commercial position is deteriorating whatever the cost report says, because the unpriced book is growing faster than it is being closed.

Code the cost as it is incurred. Each change needs cost codes that capture its cost separately from the day work starts. Costs never coded to the event cannot be isolated a year later, which is precisely when someone will need them.

7. Cash, and why an unpriced book hurts twice

Even when a variation goes well it is paid late: instructed, worked, priced, assessed, certified, then paid on ordinary terms. Until agreement, the work is being funded with nothing certified against it, so a growing unpriced book deepens the funding trough exactly when base scope is at peak spend. See `BPG-13 — Cash flow forecasting`.

8. How this goes wrong

Working on an instruction that does not exist. Verbal direction, a marked-up drawing, an email that reads like a request. Six months later the question is not whether the work was done but whether it was instructed by someone entitled to instruct it.

Missing a notice period for administrative reasons. The entitlement may be real, the cost evidenced, and the outcome still nil. Notice discipline costs almost nothing and protects everything.

Pricing the direct works and forgetting the time. Prolongation, disruption and extended preliminaries are separate heads with separate evidence. A variation "priced in full" that ignored the two-week delay it caused is priced in half.

Netting recovery against cost in the forecast. A forecast that shows a disputed variation at zero net cost, on the assumption of full recovery, is a forecast of a negotiation. The cost is real now; the recovery is contingent.

Dropping the variation budget into the current period. See §5 and §9. It corrupts every schedule index in the account and hides the corruption inside a legitimate-looking change.

A change register that starts at the change order. Everything instructed-but-unpriced is invisible, and the project's exposure is understated by the entire size of the unpriced book. The same register, left without an ageing column or an owner, stops being a control and becomes a filing cabinet.

No cost coding for the change. The cost is incurred inside the ordinary control accounts, becomes indistinguishable from base scope, and the eventual submission has to be reconstructed by estimate. This is also how a well-founded claim becomes a global one — see `BPG-12 — Claims and extension of time`.

Double-counting between a variation and a claim. The same disruption recovered once inside a star rate and again in a disruption claim. It is the most damaging quantum error there is, because finding one instance of it invites the assessor to distrust everything else in the submission.

9. Worked example

Illustrative figures. Currency USD; a single control account within a construction contract; rates and percentage additions are those stated in the contract for the purposes of this example; quantities remeasured on site; all values exclude any indirect tax, whose treatment varies by jurisdiction.

9.1 One variation, three pricing bases

The client instructs an additional **1,200 m²** of cladding at high level. The contract's valuation rules must decide which basis applies; the three candidate outcomes are:

(a) Contract bill rates. The bill carries a cladding item at **420.00 per m²**.

```
Value = 420.00 × 1,200 = 504,000
```

(b) A star rate derived from the bill rate and adjusted for conditions. The additional area is at height, needing mast climbers and slower installation, so a rate is built up:

Element	Rate
Materials	210.00 / m²
Labour: 1.6 h/m² at 68.00/h	`1.6 × 68.00 = 108.80` / m²
Access equipment	91.20 / m²
Direct cost	**410.00 / m²**
Site on-costs and overhead at 10 %	`0.10 × 410.00 = 41.00` / m²
Subtotal	451.00 / m²
Profit at 5 %	`0.05 × 451.00 = 22.55` / m²
Star rate	**473.55 / m²**

```
Value = 473.55 × 1,200 = 568,260
Difference from the bill rate route = 568,260 − 504,000 = 64,260
```

(c) Daywork at contract daywork rates. Recorded as executed, with the contract's stated percentage addition of 15 % on materials and plant:

```
Labour:    1,200 × 1.6 = 1,920 h at 82.00/h        = 157,440
Materials: 210.00 × 1,200 = 252,000, + 15 %         = 289,800
Plant:      91.20 × 1,200 = 109,440, + 15 %         = 125,856
Total                                               = 573,096
```

The spread. `573,096 − 504,000 = 69,096`, which is `69,096 ÷ 504,000 = 13.7 %` of the lowest figure. All three are arithmetically correct valuations of the same physical work. Which one applies is a question about the contract — whether the work is of similar character executed under similar conditions — and it is decided by reading the contract, not by preference.

Note where the productivity risk lands. Under (a) and (b) the contractor is paid a rate and absorbs the consequence if installation takes longer than assumed. Under (c) the paying party absorbs it, hour by hour, which is why daywork records must be signed the day the work is done.

9.2 Incorporating the agreed variation into the baseline

Assume basis (a) applies and **504,000** is agreed. The control account before incorporation, at the Month 6 data date:

Quantity	Value
Budget at completion (BAC)	6,000,000
Planned value (PV)	3,600,000
Earned value (EV)	3,300,000
Actual cost (AC)	3,500,000

```
CPI = EV ÷ AC = 3,300,000 ÷ 3,500,000 = 0.943
SPI = EV ÷ PV = 3,300,000 ÷ 3,600,000 = 0.917
```

The varied work is programmed to start in two months' time. The new BAC is `6,000,000 + 504,000 = 6,504,000` either way; what differs is the phasing.

Done wrongly — 120,000 of the variation's budget is credited as planned value in the current period because that is when the change order was issued:

```
PV = 3,600,000 + 120,000 = 3,720,000
SPI = 3,300,000 ÷ 3,720,000 = 0.887
```

The schedule performance index falls by `0.917 − 0.887 = 0.030` because of a filing decision. No work changed, no productivity changed, nobody was late.

Done correctly — the 504,000 is phased into the two months in which the varied work is planned:

```
PV at the data date = 3,600,000 (unchanged)
SPI = 3,300,000 ÷ 3,600,000 = 0.917 (unchanged)
```

From the month the varied work starts, planned value and earned value move together and the index measures performance on the enlarged scope, which is what it is for.

9.3 Reporting the unpriced book

Suppose that at the same data date the account also carries **780,000** of instructed-but-unpriced variations (bucket 2). The forecast for the baselined scope, on a persisting cost performance basis:

```
EAC (baselined scope) = BAC ÷ CPI = 6,000,000 ÷ 0.942857 = 6,363,636
EAC (including unincorporated change) = 6,363,636 + 780,000 = 7,143,636
Apparent variance against the original BAC = 7,143,636 − 6,000,000 = (1,143,636)
```

Reported as one number, that is a 1.14 million overrun and the conversation is about performance. Reported as two lines — 363,636 of cost performance variance and 780,000 of instructed scope not yet in the baseline — it is two conversations with two different owners, and only one of them is a controls problem.

Assumptions this example depends on. Rates and percentage additions are as stated in the contract for this example; the 1,200 m^2 is a remeasured quantity, not an estimate; the cost performance index is computed on the baselined scope only, with the cost of unincorporated variations excluded from AC and EV alike; no prolongation or disruption effect is included in any of the three valuations, and any such effect would be a separate assessment.

10. Checklist

At the event

- [ ] The change is logged on the day it arises, with its event date — not the day it is priced.
- [ ] The instruction reference and the name of the person who issued it are recorded; where no instruction exists, the direction has been confirmed in writing to the contract representative the same day.
- [ ] The notice position has been checked against the contract, and notice given or diarised.
- [ ] Cost codes for the change are open before work starts.

Pricing

- [ ] The valuation basis is the one the contract requires, and the reasoning is recorded.
- [ ] Star rates show their build-up element by element.
- [ ] Daywork sheets are signed contemporaneously by an authorised person.
- [ ] Time-related cost is assessed separately from the direct works.
- [ ] The submission does not recover the same cost twice under two heads.

Forecast

- [ ] Every change is assigned to a bucket: agreed, instructed-unpriced, or claimed/disputed.
- [ ] Cost is forecast gross; recovery is never netted off.
- [ ] The estimate at completion separates baselined scope from unincorporated change.

Baseline

- [ ] Agreed variations are time-phased to the varied work's planned dates, and the schedule is updated in the same cycle as the budget.
- [ ] The baseline change is in the change log with an approval reference.
- [ ] Performance trends are annotated where the baseline changed, so a step is not read as performance.

Log health

- [ ] One log, one owner, reviewed weekly.
- [ ] Ageing by status is reported monthly, with a threshold that triggers escalation.
- [ ] Submitted values and assessed values are both held, so the assessment gap is visible.
- [ ] Time effects are recorded in days, separately from money.

Related

- `BPG-04 — Baselining and baseline change control` — the procedure a change order runs through, and what a legitimate baseline change looks like.
- `BPG-12 — Claims and extension of time` — where a variation stops being a valuation and becomes an entitlement argument, and how not to double-count between the two.
- `BPG-08 — Earned value in practice` — why an unincorporated variation distorts every index in the account.
- `BPG-09 — Estimate at completion: choosing and defending a method` — reporting a forecast that separates performance from scope growth.
- `TPL-12 — Change order log` — the instrument, with the columns set out in §6.

Sources and standards

Valuation hierarchies, notice provisions and change procedures are set out differently in each contract and in the standard forms published by bodies such as FIDIC and the NEC family; this guide describes the general principles in our own words and reproduces no clause, wording or numbering from any of them, and asserts no position under any jurisdiction's law. The IFRS 15 constraint on variable consideration is named in §4 and described in principle only; the applicable treatment depends on the reporting framework in use. Internal references are BoK Domain 7 (Contracts, Commercial Management, BoQ, Invoicing & Revenue) and BoK Domain 5 (Cost Management & Cost Control). All figures in §9 are illustrative and were computed for this document.

Status and version

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